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Unmanned Aircraft System Aided Adaptive Video Streaming: A Joint Optimization Approach

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***Abstract*—Due to the coverage constraint of wireless base station, mobile users suffer from the unstable network connection and poor service quality, especially for the prevalent video services. As an alternative solution, an unmanned aerial vehicle (UAV) is able to reach the cell edge and serve ground users (GUs). In this paper, we extend the UAV applications to the more challenging adaptive streaming service over fading channel. Firstly, we decompose the system into different modules, and present mathematical models for each of them, including a trajectory model of the UAV, fading channels between the UAV and GUs, and video streaming utility. Secondly, we formulate the problem as a non-convex optimization problem by optimizing the UAV trajectory and transmit power allocation, jointly with transmission schedule and rate allocation for multiple users. The objective is to maximize the overall utility while guaranteeing the fairness among multiple users under the UAV energy budget and rate outage probability constraints. Thirdly, to tackle this problem, we first analyze the relationship between transmission rate and rate outage probability over fading channel, and then divide the original problem into three subproblems, which can be solved by leveraging the successive convex approximation (SCA) technique. Furthermore, an overall iterative algorithm over the three sub-problems is proposed to obtain a locally optimal solution by applying the block coordinate descent technique. Finally, through extensive experiments, we demonstrate that the proposed design can achieve almost 30% performance gain in terms of max-min streaming utility for all users, compared with other benchmark schemes.**

***Index Terms*—Adaptive video streaming, unmanned aerial vehicle, transmission rate allocation, UAV trajectory, fading channel.**

I. Introduction

A. Background and Motivation

Due to the blooming of smart devices, such as tablets and smartphones, an explosive demand for mobile video services has been experienced in the past few years. It is reported in the latest Cisco report [1] that mobile video traffic has accounted for 60% of the global mobile data traffic in 2016, and the value will reach 78% by 2021. As such, the cellular network should provide higher data rates to meet the ever-increasing capacity demand for video traffic and the increasing demand for high quality video [2], [3]. On the other hand, edge users who are far away from a base station (BS) often suffer from poor quality of experience (QoE). To address this increasing data challenge, the cellular traffic offloading technique is needed, i.e., deploying macro base stations (MBSs) with a lot of small cell base stations (SBSs) such as femto BSs, and pico BSs, which constitute heterogeneous networks (HetNets) [4].

However, the traditional offloading solutions typically deploy new SBSs, which may not be suitable for scenarios with highly dynamic traffic demand. For example, it is cost-ineffective to deploy new SBSs for video streaming with temporarily high user density. Furthermore, the deployment of massive SBSs can increase the total power consumption of the heterogeneous networks [5]. Recently, cellular offloading with the help of unmanned aerial vehicle (UAV) has provided an appealing solution to address such issues [6], where the main cost (i.e., energy and aircraft cost) is typically lower than that of deploying new SBSs. With the continuous endurance improvement and cost reduction of UAVs, UAV-enabled communications will become more important in the future wireless systems [7]. Equipped with advanced transceivers and batteries, UAVs have the ability to adjust their locations swiftly and dynamically to flexibly serve the ground users (GUs). The UAV is able to reach the cell edge to provide services to different cell-edge users with different video requirements, which can help offload the video traffic from the BS. Furthermore, the altitude of the UAV typically ranges from tens of meters (m) to about 100 m. Thus, the line-of-sight (LoS) channel is available for the GUs, which can be utilized to provide better wireless service [8]. However, the air-to-ground channel quality varies over time due to the high mobility of UAV, and highly dynamic wireless channels are prone to packet loss since the communication performance reacts strongly to link degradation [9].

On the other hand, by leveraging the advantages of hypertext transfer protocol based (HTTP) (e.g., network address translation traversal and transparent caching), HTTP adaptive video streaming is emerging as an appealing solution to live streaming services and video on demand over ever-changing communication channels [10], [11]. Specifically, the video streaming is adaptive such that the quality of the video is able to be adjusted based on the transmission rate or bandwidth between the UAV and the GUs. Such feature can be better utilized in the UAV-enabled wireless environment since the transmission rate of the air-to-ground (A2G) wireless link may vary substantially over time due to the physical mobility of the UAVs. While adaptive streaming fundamentally accom-

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modates transmission rate or bandwidth variations, optimized resource allocation is needed to improve the performance of video streaming over UAV-enabled wireless transmission. Furthermore, additional propulsion energy consumption is needed for the UAV to support the movement (e.g., flying) [12], and thus the limited on-board energy as well as communication resources of the UAV should be efficiently used to enhance the video streaming performance.

There exist three important issues in designing UAV-enabled adaptive video streaming for multiple GUs. First, the UAV trajectory (including UAV velocity and acceleration) should be designed such that the UAV is able to move closely to each GU to achieve better channel quality and provide better video service. Second, the transmission schedule for multiple GUs needs to be designed appropriately during the UAV's flight so that each user can access the video service fairly. Third, the transmit power and transmission rate allocation should be designed so that each user can access better video service with high streaming utility and low rate outage probability (i.e., the probability that the communication failure occurs) over the wireless fading channel under the limited UAV energy budget. The optimization of UAV trajectory and transmit power allocation, jointly with transmission schedule and transmission rate allocation for adaptive video streaming under the limited UAV energy budget is challenging, and has not been investigated before.

B. Related Work and Contribution

Recently, there have been studies focusing on the UAV trajectory design in order to optimize the communication performance. In [7], the authors maximized the minimum throughput over all GUs by optimizing the communication scheduling and association for all GUs jointly with the UAVs' trajectory. The energy-efficient UAV communication with single ground terminal via optimizing the UAV trajectory by taking into account the UAV's energy consumption was investigated in [12]. A new type of UAV-enabled relaying system was investigated in [13], where the UAV transmit power and UAV trajectory were jointly optimized subject to practical UAV mobility constraints. The deployment and movement of UAVs were investigated in [14] over a multi-UAV enabled Internet of Things (IoT) network for data collection, so as to minimize the power consumption of the active IoT devices. However, prior work mainly focused on energy efficient communication and throughput maximization without considering the transmission reliability over fading channel.

To enable the reliable transmission for UAV-enabled communication system, the reliability of the aerial platforms needs to be modelled, i.e., the A2G fading channel. It is observed in the recent studies that the UAV location has significant impact on the A2G link reliability. A generic A2G analytical framework was proposed in [15], which considered the relationship between the UAV location and the path loss exponent as well as multipath fading. The UAV trajectory and transmit power were jointly optimized for UAV relay networks in [16] to minimize the rate outage probability with the amplify-and-forward UAV relay. In [17], the authors proposed a wireless relay network model with the help of the unmanned aircraft system over fading channel, where the achievable rate outage probability was taken into account. However, the above work only focused on relay network without considering UAV's trajectory design and propulsion energy consumption.

Recently, there have been some works on energy-efficient video transmission over unstable wireless conditions. Specifically, in [18], an energy efficient data transmission scheme was proposed in mobile cloud computing by managing data for delay-tolerant or prefetching-friendly applications. An application-layer energy efficient transmission protocol was proposed in [19] to manage data transmissions for mobile apps with cloud computing. The benefits of collaborative caching in wireless streaming services was explored in [20]–[22], and both challenges of incentives and truthfulness of selfish wireless service providers were investigated. By taking into account both the spatial and temporal structures in a video, a spatial-temporal attentional mechanism was proposed in [23] and vast quantity of web images with complex backgrounds were analyzed in [24], [25]. QoE should be included in the design of adaptive video streaming systems [26]. In [27], the authors proposed an optimization framework for video streaming in heterogeneous cellular networks, via jointly optimizing time-domain resource partitioning, and rate allocation, as well as video quality selection. The green video delivery in heterogeneous cellular networks was studied in [28], where the trade-off between the network energy cost and the user perceived quality of video was analyzed. The work in [29]–[31] formulated a utility maximization problem for adaptive streaming in mobile cellular networks and showed that the utility maximization can be achieved through an efficient algorithm. However, the above work investigated the traditional cellular network without UAV communication.

In this paper, we consider the optimization problem of UAV trajectory design and transmit power allocation, jointly with transmission schedule and transmission rate allocation among multiple GUs for adaptive video streaming over fading channel. Our aim is to maximize the minimum streaming utility for all GUs, under the rate outage probability constraint over the fading channel and UAV total energy constraint, including both propulsion energy and communication related energy. To the best of our knowledge, we are the first to analyze the resource allocation for UAV-enabled adaptive video streaming over fading channel. Our preliminarily work provides a general framework to UAV-enabled video streaming, and can guide the video applications over UAV-enabled wireless systems. The main contributions are summarized as follows:

- Firstly, we derive the rate outage probability for the A2G fading channel and video streaming utility function for each GU. Next, we consider the optimization of UAV trajectory and transmit power allocation, jointly with transmission schedule and transmission rate allocation for adaptive video streaming, where the problem is formulated as a non-convex programming.
- Secondly, we analyze the relationship between the transmission rate and rate outage probability over fading channel, and find that the transmission rate allocation can be transformed to the UAV trajectory design and transmit

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power allocation with the target rate outage probability, based on which we divide the original problem into three subproblems, i.e., transmission schedule optimization problem, transmit power optimization problem, as well as UAV trajectory optimization problem.

- Thirdly, we propose an overall iterative algorithm over the three subproblems to obtain a locally optimal solution by applying the successive convex approximation (SCA) and block coordinate descent technique. Furthermore, the convergence and computation complexity of the proposed algorithm are analyzed and we also derive an upper bound for max-min streaming utility among all GUs.
- Finally, we evaluate the performance of the proposed solution together with other benchmark schemes and the derived upper bound. Simulations are conducted and the results reveal that the proposed solution outperforms other benchmark schemes in terms of max-min streaming utility among all GUs.

C. Organization and Notation

The remainder of this paper is organized as follows. Section II describes our system model and formulates the joint design problem with a mixed-integer non-convex programming, which is difficult to solve in general. Section III divides the original problem into three subproblems, and proposes an overall iterative algorithm over the three subproblems to obtain a locally optimal solution by leveraging the SCA and block coordinate descent technique. The convergence of the proposed algorithm and the upper bound of the performance are analyzed in Section IV. In Section V, we present simulation results, and we conclude the paper in Section VI.

In this paper, we denote the scalars with italic letters, and use boldface letters to denote the vectors. We use $\mathbb{R}^{M\times 1}$ to denote the space of M-dimensional real-valued vectors, and $\|\mathbf{a}\|$ and $\mathbf{a}^T$ represent the Euclidean norm and transpose of vector $\mathbf{a}$, respectively. $\log(\cdot)$ and $\log_2(\cdot)$ are used to denote the logarithm with base e and 2, respectively. $|M|$ denotes the cardinality of set M and $\mathbb{P}(\cdot)$ denotes the probability.

II. System Model and Problem Statement

In this section, we will introduce the system model for the UAV-enabled adaptive video streaming and the corresponding problem formulation. We summarize the key notations used in this paper in Table I.

A. System Model

We consider a wireless video service system where a fixed-wing UAV is employed as a base station to deliver videos to multiple GUs, where the requested videos are stored at the UAV, as shown in Fig. 1, since fixed-wing UAV consumes less propulsion energy than rotary-wing UAV [17]. The scenario can also be extended to the case of multiple UAVs by assigning different UAVs to serve different groups of GUs as in [14]. Assume that there are N GUs, denoted by $u_1, u_2, \ldots, u_N$. The location of each user u_n is given by $\mathbf{w}_n$, where $\mathbf{w}_n \in \mathbb{R}^{2\times 1}$ represents the horizontal coordinate of u_n on the ground,

TABLE I
NOTATION AND DEFINITION

Notation	Definition
B	Signal bandwidth (Hz)
H	The altitude of UAV (m)
N	Number of ground users
K_c	Rician fading factor
$P_{\max}$	The maximum transmit power of UAV (W)
T	Number of time slots for the UAV operation
δ	The elemental time slot length (s)
$V_{\max}$	The maximum speed of UAV (m/s)
$V_{\min}$	The minimum speed of UAV (m/s)
$a_{\max}$	The maximum acceleration of UAV (m^2/s)
u_n	The n-th GU
$\mathbf{w}_n$	(2D) coordinate of u_n
$\mathbf{q}[t]$	The UAV's horizontal location at time slot t
$\mathbf{v}[t]$	The UAV's velocity at time slot t
$\mathbf{a}[t]$	The UAV's acceleration at time slot t
$P[t]$	UAV transmit power at time slot t (W)
$p_n^{out}[t]$	Outage probability for u_n at time slot t
$R_n[t]$	UAV transmission rate for u_n at time slot t (bps)
$x_n[t]$	Scheduling variable for u_n at time slot t
$\beta_n[t]$	Large-scale channel coefficient for u_n at time slot t
$\rho_n[t]$	Small-scale channel coefficient for u_n at time slot t
$U_n[t]$	Streaming utility for u_n at time slot t
$E_{\max}$	The total UAV energy budget (Joule)

$1 \leq n \leq N$. The N GUs are waiting to be scheduled for communication by the UAV for HTTP adaptive video streaming.

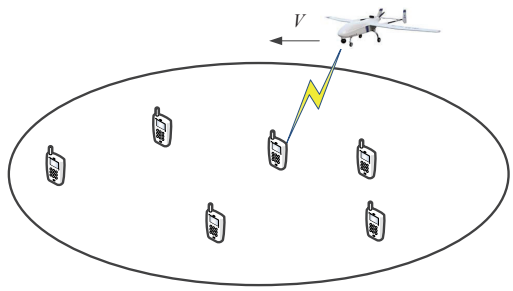

Fig. 1. System model for the UAV-enabled wireless system

The initial and final locations of the UAV are assumed to be pre-determined, which are denoted as $\mathbf{w}_0, \mathbf{w}_F \in \mathbb{R}^{2\times 1}$, respectively. $\mathbf{w}_0$ and $\mathbf{w}_F$ can be determined, for example, by the UAV's launching or landing locations [13]. Furthermore, the UAV is assumed to fly at a fixed altitude H (m) with a given time duration, during which the UAV serves the GUs via a time-division multiple access (TDMA) scheme. Assume that time duration consists of $T > 0$ time slots, with δ denoting the elemental length. Note that δ can be set to be small enough such that the distances between the UAV and all GUs are approximately unchanged within each time slot [13]. As a result, the UAV's flying trajectory projected on the ground can be approximated by the set $\{\mathbf{q}[t], 0 \leq t \leq T\}$, where $\mathbf{q}[t] \in \mathbb{R}^{2\times 1}$ denotes the UAV's horizontal location at time slot t. Furthermore, we have $\mathbf{q}[0] = \mathbf{w}_0$ and $\mathbf{q}[T] = \mathbf{w}_F$.

Let $\mathbf{v}[t] \in \mathbb{R}^{2\times 1}$ and $\mathbf{a}[t] \in \mathbb{R}^{2\times 1}$ be the velocity and acceleration of the UAV at time slot t, respectively. Furthermore, we define $\mathbf{v}_0$ as the initial velocity of the UAV, and then $\mathbf{v}[0] = \mathbf{v}_0$. Thus, for all $0 \leq t \leq T-1$, we have

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$\mathbf{v}[t+1]-\mathbf{v}[t]=\mathbf{a}[t]\delta$ and $\mathbf{q}[t+1]-\mathbf{q}[t]=\mathbf{v}[t]\delta+\frac{1}{2}\mathbf{a}[t]\delta^2$. As such, it can be verified that given UAV trajectory $\{\mathbf{q}[t],\forall t\}$ and $\mathbf{v}_0$, the UAV velocity $\{\mathbf{v}[t],\forall t\}$ and acceleration $\{\mathbf{a}[t],\forall t\}$ can be determined uniquely. In the following, we refer that the UAV trajectory consists of the UAV velocity and acceleration.

Let $V_{\max}$ and $a_{\max}$ be the maximum UAV speed and acceleration, and $V_{\min}$ denotes the minimum speed required for the fixed-wing UAV to remain aloft. Then, we have $V_{\min} \leq \|\mathbf{v}[t]\| \leq V_{\max}$, $\|\mathbf{a}[t]\| \leq a_{\max}, \forall t$. As derived in [12], the propulsion energy consumption of the fixed-wing UAV measured in Joule can be approximately expressed as

$$E_p \approx \sum_{t=1}^{T} \delta \left(c_1 \|\mathbf{v}[t]\|^3 + \frac{c_2}{\|\mathbf{v}[t]\|} \left(1 + \frac{\|\mathbf{a}[t]\|^2}{g^2} \right) \right), \quad (1)$$

where c_1 and c_2 are constants which are related to air density, drag coefficient, wing area etc, g denotes the gravitational acceleration with value 9.8 m/s^2.

B. Channel Model

It is assumed that the instantaneous channel state information (CSI) of the A2G channel is unavailable at the UAV and only the channel distribution information (CDI) is available at the UAV. Furthermore, we assume quasi-static fading channel for the A2G channel, where the instantaneous channel coefficient between the UAV and GU remain unchanged for each time slot, and may vary across different time slots. To this end, the channel coefficient between the UAV and GU u_n at time slot t can be modelled as $h_n[t] = \rho_n[t]\sqrt{\beta_n[t]}, \forall n, t$, where $\rho_n[t]$ is a complex-valued random variable accounting for small-scale fading, $\beta_n[t]$ accounts for the large-scale channel attenuation, which depends on the distance between the UAV and GU. Specifically, we denote $d_n[t]$ as the distance between the UAV and u_n at time slot t, then

$$\beta_n[t] = \beta_0 d_n^{-\alpha}[t] = \frac{\beta_0}{(H^2 + \|\mathbf{q}[t] - \mathbf{w}_n\|^2)^{\alpha/2}}, \quad (2)$$

where β_0 denotes the channel power gain at $d = 1$m, and $\alpha \geq 2$ is the path loss exponent.

We assume that the small-scale fading of the A2G channel follows Rician fading model since the A2G channel typically consists of a LoS link. In the Rician fading channel, the Rician factor K_c represents for the power ratio between the LoS signal component and the scattered component. As a result, similar to that in [17], the cumulative distribution function (CDF) of $|\rho_n[t]|^2$ is obtained as follows, i.e.,

$$\begin{aligned} F(z) &\triangleq \mathbb{P}(|\rho_n[t]|^2 < z) \\ &= 1 - Q_1(\sqrt{2K_c}, \sqrt{2(K_c+1)z}), \end{aligned} \quad (3)$$

where $Q_1(a,b)$ denotes the Marcum-Q function with modified Bessel function I_0 of order 0,

$$Q_1(a,b) = \int_b^{\infty} x e^{-\frac{x^2+a^2}{2}} I_0(ax) dx. \quad (4)$$

.

Define scheduling variable $x_n[t]$ to denote whether UAV decides to transmit video data to GU u_n at time slot t or not. If u_n is served at time slot t, then $x_n[t] = 1$, otherwise, $x_n[t] = 0$. Let $R_n[t]$ denote the transmission rate for u_n at time slot t, $\forall n, t$, which are design variables. We denote the UAV transmit power at time slot t as $P[t]$, which are also design variables. Thus, $0 \leq P[t] \leq P_{\max}$, where $P_{\max}$ is the maximum allowable transmit power of the UAV. As a result, the total communication related energy consumption of the UAV is given by

$$E_c = \sum_{t=1}^{T} \delta P[t]. \quad (5)$$

The instantaneous channel capacity between the UAV and GU u_n at time slot t measured in bits/second (bps) is thus given by

$$\begin{aligned} C_n[t] &\triangleq B \log_2 \left(1 + \frac{P[t]|h_n[t]|^2}{\sigma^2} \right) \\ &= B \log_2 \left(1 + \frac{P[t]\beta_n[t]|\rho_n[t]|^2}{\sigma^2} \right), \end{aligned} \quad (6)$$

where B is the signal bandwidth measured in Hertz (Hz), and σ^2 denotes the noise power at the GUs.

Similar as in [17], we define the rate outage probability $p_n^{out}[t]$ as the probability that the communication between the UAV and GU u_n fails at time slot t, which can be obtained from

$$\begin{aligned} p_n^{out}[t] &\triangleq \mathbb{P}(C_n[t] < R_n[t]) \\ &= \mathbb{P}\left(|\rho_n[t]|^2 < \frac{2^{R_n[t]/B} - 1}{P[t]\beta_n[t]/\sigma^2} \right) \\ &= F\left(\frac{(2^{\frac{R_n[t]}{B}} - 1)(H^2 + \|\mathbf{q}[t] - \mathbf{w}_n\|^2)^{\alpha/2}}{P[t]\gamma_0} \right), \end{aligned} \quad (7)$$

where $\gamma_0 \triangleq \beta_0/\sigma^2$, and the last equality holds due to the CDF definition in (3). In the following, we refer to the outage probability as the rate outage probability defined in (7).

C. Adaptive Video Streaming Model

Video files are delivered to multiple GUs through the UAV system, which has been described in the system model. We assume that HTTP adaptive video streaming is used for video delivery. To facilitate the analysis, we consider a simplified video streaming utility model, in which the user's utility only relates to the video transmission rate. Intuitively, if a user can receive higher video rate, then it can perceive higher quality of experience. Denote $U(\cdot) : \mathbb{R} \to \mathbb{R}$ as a non-decreasing function which maps video transmission rate $R_n[t]$ to video quality $U_n[t] \triangleq U(R_n[t])$ perceived by GU u_n at time slot t.

Note that the user experience is typically not linearly proportional to the video rate, since the user experience tends to saturate at the higher video rates. In fact, the user experience follows the logarithmic law [32], and thus the streaming utility function can be modelled in the logarithmic form for video streaming. As a result, we adopt the video streaming utility model as the logarithmic function of video transmission rate in [29], [33] as follows:

$$U_n[t] = \theta \log \frac{R_n[t]}{r_n} \quad (8)$$

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where the constants θ is positive, which may be different for various applications. r_n denotes the required playback rate for GU u_n, which depends on the physical capability of the user's device (e.g., screen size). Note that the streaming utility function $U(\cdot)$ is positive, concave with respect to $R_n[t]$, continuous and twice differentiable for $R_n[t]$.

Therefore, the total streaming utility for user u_n can be expressed as $\delta \sum_{t=1}^{T} x_n[t]U_n[t]$ during T time slots. In order to give a maximum protection to the GU with minimum QoE, we choose to provide the max-min fairness of video streaming service among all GUs. Therefore, the objective is to maximize the minimum streaming utility among all GUs, i.e., $\min_n \delta \sum_{t=1}^{T} x_n[t]U_n[t]$ needs to be maximized.

D. Problem Formulation

Let $\mathbf{X} \triangleq \{x_n[t], \forall n, t\}$, $\mathbf{R} \triangleq \{R_n[t], \forall n, t\}$, $\mathbf{P} \triangleq \{P[t], \forall t\}$, $\mathbf{Q} \triangleq \{\mathbf{q}[t], \forall t\}$, $\mathbf{V} \triangleq \{\mathbf{v}[t], \forall t\}$, $\mathbf{A} \triangleq \{\mathbf{a}[t], \forall t\}$. Our objective is to maximum the minimum streaming utility among all users by optimizing the UAV trajectory $\mathbf{Q}$ (including UAV velocity $\mathbf{V}$ and acceleration $\mathbf{A}$) and transmit power $\mathbf{P}$, joint with transmission schedule $\mathbf{X}$ and transmission rate $\mathbf{R}$, under the UAV total energy budget constraint and the outage probability constraint of the A2G channel that the outage probability for each GU belows a target threshold ϵ. The outage probability for each GU u_n at time slot t can be expressed as $x_n[t]p_n^{out}[t]$, then we have $x_n[t]p_n^{out}[t] \leq \epsilon, \forall n, t$. On the other hand, the transmission outage probability over fading channel will cause video playout to stall and consequently degrade QoE. The total streaming utility for user u_n can thus be approximated by $\delta\theta \sum_{t=1}^{T} x_n[t] \log \frac{R_n[t](1-\epsilon)}{r_n}$. The optimization problem is formulated as

$$
\begin{aligned}
\text{(P1)}: \max_{\mathbf{X},\mathbf{R},\mathbf{P},\mathbf{Q},\mathbf{V},\mathbf{A}} \quad & \min_n \delta\theta \sum_{t=1}^{T} x_n[t] \log \frac{R_n[t](1-\epsilon)}{r_n} \\
\text{s.t.} \quad & x_n[t]p_n^{out}[t] \leq \epsilon, \forall n, t, && (9)\\
& x_n[t] \in \{0,1\}, \quad \forall n, t, && (10)\\
& \sum_{n=1}^{N} x_n[t] \leq 1, \quad \forall t, && (11)\\
& E_p + E_c \leq E_{\max}, && (12)\\
& V_{\min} \leq \|\mathbf{v}[t]\| \leq V_{\max}, \forall t, && (13)\\
& \mathbf{v}[t+1] - \mathbf{v}[t] = \mathbf{a}[t]\delta, \forall t, && (14)\\
& \mathbf{q}[t+1] - \mathbf{q}[t] = \mathbf{v}[t]\delta + \frac{1}{2}\mathbf{a}[t]\delta^2, \forall t, && (15)\\
& \mathbf{q}[0] = \mathbf{w}_0, \mathbf{q}[T] = \mathbf{w}_F, \mathbf{v}[0] = \mathbf{v}_0, && (16)\\
& 0 \leq P[t] \leq P_{\max}, \forall t, && (17)
\end{aligned}
$$

where $E_{\max}$ denotes the UAV total energy budget, including both propulsion energy and communication related energy.

In the formulation, the objective is to maximize the minimum streaming utility among all GUs. The constraints in (13)-(16) represent the UAV velocity and acceleration constraints and initial/final location constraints of the UAV, respectively. Problem (P1) is a mixed-integer non-convex optimization problem due to the binary constraints in (10) and non-convex constraints in (9), (12) and (13), as well as the non-convex objective function. Furthermore, (P1) contains the coupled optimization variables and the complicated representation of streaming utility function and outage probability. Therefore, the mixed-integer non-convex problem (P1) is difficult to be solved optimally within polynomial time.

III. Proposed Solution

In this section, we first analyze the relationship between transmission rate and outage probability, and find that the transmission rate allocation can be transformed to the UAV trajectory design and transmit power allocation with the target outage probability. Next, we divide the original problem into three subproblems, and then propose an overall iterative algorithm over the three subproblems to obtain a locally optimal solution that satisfies the the Karush-Kuhn-Tucker (KKT) conditions by applying the SCA and block coordinate descent technique. Although SCA and block coordinate descent technique were also adopted in related works, e.g., [13], [34], the investigated problems in these works are fundamentally different from our problem.

Define $\mu \triangleq \min_n \delta\theta \sum_{t=1}^{T} x_n[t] \log \frac{R_n[t](1-\epsilon)}{r_n}$, then problem (P1) can be simplified as follows:

$$
\begin{aligned}
\text{(P2)}: \max_{\mathbf{X},\mathbf{R},\mathbf{P},\mathbf{Q},\mathbf{V},\mathbf{A},\mu} \quad & \mu \\
\text{s.t.} \quad & x_n[t]p_n^{out}[t] \leq \epsilon, \forall n, t, && (18)\\
& \delta\theta \sum_{t=1}^{T} x_n[t] \log \frac{R_n[t](1-\epsilon)}{r_n} \geq \mu, \forall n, && (19)\\
& (10) - (17).
\end{aligned}
$$

By revising the constraint in (18), we can obtain the problem (P3) and the following theorem.

$$
\begin{aligned}
\text{(P3)}: \max_{\mathbf{X},\mathbf{R},\mathbf{P},\mathbf{Q},\mathbf{V},\mathbf{A},\mu} \quad & \mu \\
\text{s.t.} \quad & p_n^{out}[t] = \epsilon, \forall n, t, && (20)\\
& (10) - (17), (19).
\end{aligned}
$$

Theorem 1: Problem (P2) is equivalent to problem (P3).

Proof It can be verified that without loss of optimality to problem (P2), constraints in (18) can always be replaced by constraints in (20). Otherwise, if at the optimal solution to (P2), there exists n, t that constraint in (18) is satisfied while the corresponding constraint in (20) is not satisfied. Then, we can always increase $R_n[t]$ with other variables fixed to make constraint in (20) be satisfied without changing the objective value, since $p_n^{out}[t]$ is a non-decreasing function with respect to $R_n[t]$. As a result, there always exists an optimal solution to (P2) such that the constraints in (20) are satisfied. Therefore, solving problem (P2) is equivalent to solving problem (P3), and then the proof is completed. □

By substituting $p_n^{out}[t] = \epsilon$ with (7), (P3) can be equivalently written as

$$
\begin{aligned}
\text{(P4)}: \max_{\mathbf{X},\mathbf{P},\mathbf{Q},\mathbf{V},\mathbf{A},\mu} \quad & \mu \\
\text{s.t.} \quad & (10) - (17), (19),
\end{aligned}
$$

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$$R_n[t] = B\log_2\left(1+\frac{F^{-1}(\epsilon)P[t]\gamma_0}{(H^2+\|\mathbf{q}[t]-\mathbf{w}_n\|^2)^{\alpha/2}}\right), \forall n,t, \quad (21)$$

where $F^{-1}(\cdot)$ is the inverse function of $F(\cdot)$. Therefore, the transmission rate allocation can be transformed to the UAV trajectory design and transmit power allocation with the target outage probability.

Note that no closed-form expression for $F^{-1}(\epsilon)$ exists since the computation of the Marcum-Q function is complicated. However, it can be verified that $F^{-1}(\epsilon)$ is a non-decreasing function since $F(\cdot)$ is a non-decreasing function, and $0 \leq \epsilon \leq 1$. Given ϵ, the value for $F^{-1}(\epsilon)$ can thus be obtained efficiently with the bisection numerical method similar as in [35]. Given $F^{-1}(\epsilon)$, (P4) is still a mixed-integer non-convex optimization problem. To tackle such issues, the binary variables in (10) can be first relaxed into continuous variables, and then the problem is written as

$$\begin{aligned}(\text{P5}): \max_{\mathbf{X},\mathbf{P},\mathbf{Q},\mathbf{V},\mathbf{A},\mu} \quad & \mu \\ \text{s.t.} \quad & (11)-(17),(19),(21), \\ & 0 \leq x_n[t] \leq 1, \forall n,t. \end{aligned} \quad (22)$$

Note that with such a relaxation, we can reconstruct the binary solution easily by further dividing each time slot into multiple sub-slots until zero relaxation gap is achieved by the binary solution similar as in [7]. However, although relaxed, problem (P5) is still non-convex and difficult to solve directly, due to the non-convex constraints in (12), (13) as well as (19) with coupled variables $\mathbf{X}$, $\mathbf{P}$ and $\mathbf{Q}$.

In the following subsections, an efficient sub-optimal solution to (P5) is proposed by applying SCA and block coordinate descent technique. The key idea is to solve the three sub-problems of (P5) iteratively, namely *transmission schedule optimization with fixed transmit power and trajectory*, and *transmit power optimization with fixed transmission schedule and trajectory*, as well as *trajectory optimization with fixed transmission schedule and transmit power*. Then, we employ the block coordinate descent method to optimize the three sets of variables in an alternating manner until the objective value converges.

A. Transmission Schedule Optimization with Fixed Transmit Power and Trajectory

First, with any given transmit power allocation $\mathbf{P}$ and UAV trajectory $\mathbf{Q}$, we consider the sub-problem to optimize the transmission schedule $\mathbf{X}$ only. In this case, the total energy consumption of the UAV $E_p + E_c$ and $R_n[t]$ are fixed due to (1), (5) and (21). This sub-problem of (P5) is given by

$$\begin{aligned}(\text{P6}): \max_{\mathbf{X},\mu} \quad & \mu \\ \text{s.t.} \quad & \sum_{n=1}^{N} x_n[t] \leq 1, \qquad \forall t, \\ & \delta\theta\sum_{t=1}^{T} x_n[t]\log\frac{R_n[t](1-\epsilon)}{r_n} \geq \mu, \forall n, \quad (23) \\ & 0 \leq x_n[t] \leq 1, \forall n,t. \quad (24)\end{aligned}$$

Note that problem (P6) is a standard linear programming (LP) problem since all constraints and the objective function are linear. Therefore, (P6) can be solved efficiently by standard LP techniques within polynomial time.

B. Transmit Power Optimization with Fixed Transmission Schedule and Trajectory

In this subsection, with any given transmission schedule $\mathbf{X}$ and UAV trajectory $\mathbf{Q}$, we consider the sub-problem of (P5) to optimize the UAV transmit power allocation $\mathbf{P}$. In this case, E_p is fixed due to (1). This sub-problem can be written as

$$\begin{aligned}(\text{P7}): \max_{\mathbf{P},\mu} \quad & \mu \\ \text{s.t.} \quad & 0 \leq P[t] \leq P_{\max}, \quad (25) \\ & \sum_{t=1}^{T}\delta P[t] + E_p \leq E_{\max}, \quad (26) \\ & \delta\theta\sum_{t=1}^{T} x_n[t]\log\frac{R_n[t](1-\epsilon)}{r_n} \geq \mu, \forall n, \quad (27) \\ & R_n[t] = B\log_2\left(1+\frac{F^{-1}(\epsilon)P[t]\gamma_0}{(H^2+\|\mathbf{q}[t]-\mathbf{w}_n\|^2)^{\alpha/2}}\right), \forall n,t. \quad (28)\end{aligned}$$

It can be shown that $R_n[t]$ is a concave function with respect to $P[t]$, and then the left-hand sides (LHSs) of constraints in (27) are now concave with respect to $P[t]$ due to the composition rules that preserves concavity. Therefore, problem (P7) is a convex optimization problem, which can be solved efficiently by standard convex optimization techniques or solvers such as CVX [36].

C. Trajectory Optimization with Fixed Transmission Schedule and Transmit Power

In this subsection, for any given transmission schedule $\mathbf{X}$ and UAV transmit power allocation $\mathbf{P}$, we consider sub-problem of (P5) to optimize the UAV trajectory $\mathbf{Q}$ (including UAV velocity $\mathbf{V}$ and acceleration $\mathbf{A}$). In this case, E_c is fixed due to (5). This sub-problem can be written as

$$\begin{aligned}(\text{P8}): \max_{\mathbf{Q},\mathbf{V},\mathbf{A},\mu} \quad & \mu \\ \text{s.t.} \quad & V_{\min} \leq \|\mathbf{v}[t]\| \leq V_{\max}, \forall t, \quad (29) \\ & \delta\theta\sum_{t=1}^{T} x_n[t]\log\frac{R_n[t](1-\epsilon)}{r_n} \geq \mu, \forall n, \quad (30) \\ & E_c + \sum_{t=1}^{T}\delta\left(c_1\|\mathbf{v}[t]\|^3 + \frac{c_2}{\|\mathbf{v}[t]\|} + \frac{c_2\|\mathbf{a}[t]\|^2}{\|\mathbf{v}[t]\|\, g^2}\right) \leq E_{\max}, \quad (31) \\ & R_n[t] = B\log_2\left(1+\frac{F^{-1}(\epsilon)P[t]\gamma_0}{(H^2+\|\mathbf{q}[t]-\mathbf{w}_n\|^2)^{\alpha/2}}\right), \forall n,t, \quad (32) \\ & \mathbf{v}[t+1]-\mathbf{v}[t] = \mathbf{a}[t]\delta, \forall t, \quad (33) \\ & \mathbf{q}[t+1]-\mathbf{q}[t] = \mathbf{v}[t]\delta + \frac{1}{2}\mathbf{a}[t]\delta^2, \forall t, \quad (34) \\ & \mathbf{q}[0] = \mathbf{w}_0, \mathbf{q}[T] = \mathbf{w}_F, \mathbf{v}[0] = \mathbf{v}_0, \quad (35)\end{aligned}$$

Note that problem (P8) is non-convex since non-convex constraints exist in (29) to (31). In the following, we first

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introduce the slack variables $\mathbf{O} \triangleq \{o[t], \forall t\}$, and then we have

$$
\begin{aligned}
(\text{P9}): \max_{\mathbf{Q},\mathbf{V},\mathbf{A},\mathbf{O},\mu} \quad & \mu \\
\text{s.t.} \quad & (32)-(35), \\
& \|\mathbf{v}[t]\|^2 \le V_{\max}^2, \forall t, && (36)\\
& \|\mathbf{v}[t]\|^2 \ge V_{\min}^2, \forall t, && (37)\\
& \|\mathbf{v}[t]\|^2 \ge o[t]^2, \forall t, && (38)\\
& o[t] \ge 0, \forall t, && (39)\\
& \delta\theta \sum_{t=1}^{T} x_n[t] \log \frac{R_n[t](1-\epsilon)}{r_n} \ge \mu, \forall n, && (40)\\
& E_c + \sum_{t=1}^{T} \delta \left(c_1 \|\mathbf{v}[t]\|^3 + \frac{c_2}{o[t]} + \frac{c_2 \|\mathbf{a}[t]\|^2}{o[t] g^2} \right) \le E_{\max}. && (41)
\end{aligned}
$$

Theorem 2: Problem (P8) is equivalent to problem (P9).

Proof It can be shown that without loss of optimality to problem (P9), we always have $o[t] = \|\mathbf{v}[t]\|, \forall t$ in (38). Otherwise, $o[t]$ can be increased with other variables fixed to obtain another feasible solution with the same objective value. Therefore, there always exists an optimal solution to (P9) such that all constraints in (38) are satisfied with equality. On the other hand, all constraints in (29) are equivalent to constraints in (36) and (37). Therefore, problem (P8) is equivalent to problem (P9), which leads to the desired result. □

With such a reformulation, the LHS of constraint in (41) is now jointly convex with respect to $\mathbf{V}$, $\mathbf{A}$ and $\mathbf{O}$. To tackle the non-convex constraints in (37), (38) and (40), the SCA technique can be applied where the original function is approximated by a more tractable function at a given local point in each iteration. Specifically, for the constraints in (37) and (38), since $\|\mathbf{v}[t]\|^2$ is convex with respect to $\mathbf{v}[t]$, then we have the following inequality by applying the first-order Taylor expansion at the given point $\mathbf{v}^r[t]$ in the r-th iteration,

$$\|\mathbf{v}[t]\|^2 \ge \|\mathbf{v}^r[t]\|^2 + 2(\mathbf{v}^r[t])^T(\mathbf{v}[t]-\mathbf{v}^r[t]), \forall t. \quad (42)$$

For the constraints in (40), with given local point $\mathbf{q}^r[t]$ in the r-th iteration, we can also obtain the global lower bound for $R_n[t]$ by applying the first-order Taylor expansion as in [35], i.e.,

$$
\begin{aligned}
R_n[t] &= B \log_2 \left(1 + \frac{P[t]\gamma_0}{H^2 + \|\mathbf{q}[t]-\mathbf{w}_n\|^2} \right) && (43)\\
&\ge A_{n,r}[t] - I_{n,r}[t] \left(\|\mathbf{q}[t]-\mathbf{w}_n\|^2 - \|\mathbf{q}^r[t]-\mathbf{w}_n\|^2 \right) \triangleq R_n^{lb}[t],
\end{aligned}
$$

where

$$
\begin{aligned}
A_{n,r}[t] &= B \log_2 \left(1 + \frac{F^{-1}(\epsilon)P[t]\gamma_0}{J_{n,r}[t]^{\alpha/2}} \right),\\
I_{n,r}[t] &= \frac{BF^{-1}(\epsilon)P[t]\gamma_0(\alpha/2)\log_2 e}{J_{n,r}[t]\left(J_r[t]^{\alpha/2} + F^{-1}(\epsilon)P[t]\gamma_0\right)},\\
J_{n,r}[t] &= H^2 + \|\mathbf{q}^r[t]-\mathbf{w}_n\|^2. && (44)
\end{aligned}
$$

With any given local points $\mathbf{V}^r \triangleq \{\mathbf{v}^r[t], \forall t\}, \mathbf{Q}^r \triangleq \{\mathbf{q}^r[t], \forall t\}$ and the lower bound expressions in (42) and (43), problem (P9) is approximated as the following (P10), where the efficiency of approximation is also evaluated in Section V.

$$
\begin{aligned}
(\text{P10}): \max_{\mathbf{Q},\mathbf{V},\mathbf{A},\mathbf{O},\mu} \quad & \mu \\
\text{s.t.} \quad & (33)-(35), \\
& \|\mathbf{v}[t]\|^2 \le V_{\max}^2, \forall t, && (45)\\
& \|\mathbf{v}^r[t]\|^2 + 2(\mathbf{v}^r[t])^T(\mathbf{v}[t]-\mathbf{v}^r[t]) \ge V_{\min}^2, \forall t, && (46)\\
& \|\mathbf{v}^r[t]\|^2 + 2(\mathbf{v}^r[t])^T(\mathbf{v}[t]-\mathbf{v}^r[t]) \ge o[t]^2, \forall t, && (47)\\
& o[t] \ge 0, \forall t, && (48)\\
& \delta\theta \sum_{t=1}^{T} x_n[t] \log \frac{R_n^{lb}[t](1-\epsilon)}{r_n} \ge \mu, \forall n, && (49)\\
& E_c + \sum_{t=1}^{T} \delta \left(c_1 \|\mathbf{v}[t]\|^3 + \frac{c_2}{o[t]} + \frac{c_2 \|\mathbf{a}[t]\|^2}{o[t] g^2} \right) \le E_{\max}. && (50)
\end{aligned}
$$

It can be shown that $R_n^{lb}[t]$ is a concave function with respect to $\mathbf{q}[t]$, then the LHSs of constraints in (49) are concave with respect to $\mathbf{q}[t]$ due to the the composition rules that preserves concavity. The LHSs of constraints in (46) and (47) are concave with respect to $\mathbf{v}[t]$, and all other constraints are convex constraints. Therefore, problem (P10) is a convex optimization problem, which can be solved efficiently by standard convex optimization techniques or solvers such as CVX.

Note that due to the lower bounds given in (42) and (43), if the constraints in (46), (47) and (49) of (P10) are satisfied, then the constraints in (37), (38) and (40) of (P9) with the same $\{\mathbf{Q}, \mathbf{V}, \mathbf{A}, \mathbf{O}, \mu\}$ are also guaranteed to be satisfied, but the reverse is not true. As a result, the feasible region of (P10) is a subset of that for (P9) in general, and thus the optimal value of (P10) can be regarded as a lower bound to that of (P9) with fixed transmission schedule and transmit power allocation.

D. Overall Iterative algorithm

Based on the above results, the original problem (P1) can be solved by applying the SCA and block coordinate descent technique [34]. Specifically, the optimization variables are divided into three blocks, i.e., $\{\mathbf{X}, \mathbf{P}, \mathbf{Q}\}$. The transmission schedule $\mathbf{X}$, as well as transmit power allocation $\mathbf{P}$ and UAV trajectory $\mathbf{Q}$ are alternately optimized, via solving problem (P6), (P7) or (P10) in each iteration correspondingly. We summarize the details of the alternating optimization method in the following Algorithm 1.

IV. Discussions

In this section, we first analyze the convergence and complexity of Algorithm 1, then we will give an upper bound of the max-min streaming utility of our proposed design.

For ease of exposition, we define $\mu(\mathbf{X}^r, \mathbf{P}^r, \mathbf{Q}^r)$ and $\mu_{lb}(\mathbf{X}^r, \mathbf{P}^r, \mathbf{Q}^r)$ as the objective value of (P7) and (P10) with given $\mathbf{X}^r$, $\mathbf{P}^r$, and $\mathbf{Q}^r$ over r-th iteration, respectively. It then follows that

$$\mu(\mathbf{X}^r, \mathbf{P}^r, \mathbf{Q}^r) \overset{(a)}{\le} \mu(\mathbf{X}^{r+1}, \mathbf{P}^r, \mathbf{Q}^r) \overset{(b)}{\le} \mu(\mathbf{X}^{r+1}, \mathbf{P}^{r+1}, \mathbf{Q}^r)$$

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Algorithm 1 Iterative transmission schedule, transmit power and trajectory optimization for (P1)

1: Initialize the UAV trajectory and transmit power as $\mathbf{Q}^0$ and $\mathbf{P}^0$;
2: Let $r \leftarrow 0$;
3: tolerance $\kappa > 0$;
4: Obtain $F^{-1}(\epsilon)$ with bisection numerical method;
5: **repeat**
6: Solve LP problem (P6) for given $\mathbf{P}^r$ and $\mathbf{Q}^r$ to obtain the optimal solution $\mathbf{X}^{r+1}$;
7: Solve convex problem (P7) for given $\mathbf{X}^{r+1}$ and $\mathbf{Q}^r$, then denote the optimal solution as $\mathbf{P}^{r+1}$;
8: Solve convex problem (P10) for given $\mathbf{X}^{r+1}$ and $\mathbf{P}^{r+1}$ as well as $\mathbf{Q}^r$, then denote the optimal solution as $\mathbf{Q}^{r+1}$;
9: $r \leftarrow r+1$;
10: **until** The fractional increase of the objective value is below κ.

$$\overset{(c)}{\leq} \mu_{lb}(\mathbf{X}^{r+1}, \mathbf{P}^{r+1}, \mathbf{Q}^r) \overset{(d)}{\leq} \mu_{lb}(\mathbf{X}^{r+1}, \mathbf{P}^{r+1}, \mathbf{Q}^{r+1}) \overset{(e)}{\leq} \mu(\mathbf{X}^{r+1}, \mathbf{P}^{r+1}, \mathbf{Q}^{r+1}) \quad (51)$$

Note that (a) holds since the optimal solution $\mathbf{X}^{r+1}$ is obtained for given $\mathbf{P}^r$ and $\mathbf{Q}^r$ at step 6 of Algorithm 1, (b) holds since the optimal solution $\mathbf{P}^{r+1}$ is obtained with given $\mathbf{X}^{r+1}$ and $\mathbf{Q}^r$ at step 7 of Algorithm 1. At step 8 of Algorithm 1, SCA technique is leveraged to solve problem (P10). Since monotonic convergence is guaranteed in the SCA technique [7], then (c)-(e) must hold. On the other hand, the objective value of (P1) is upper bounded by a finite value. Therefore, the proposed Algorithm 1 is guaranteed to converge. Furthermore, since the lower bound functions given in (42) and (43) have the same gradients as their original functions at the given local points, the proposed Algorithm 1 is able to converge to a locally optimal solution to (P1) that satisfies the KKT conditions [7], [13].

It is shown in Algorithm 1 that the overall iterative algorithm requires alternatively solving the standard LP problem (P6), the standard convex optimization problems (P7) and (P10), each with the polynomial complexity. As a result, the time complexity of the proposed iterative Algorithm 1 is roughly given by $O(MN^3T^3)$ [37], where M is number of iterations.

Theorem 3: The max-min streaming utility among all GUs is upper bounded by $\frac{\delta\theta T}{N}\log\left(\frac{B(1-\epsilon)}{\min_n r_n}\log_2\left(1+\frac{F^{-1}(\epsilon)P_{\max}\gamma_0}{H^\alpha}\right)\right)$.

Proof Consider the ideal case that the total energy budget of the UAV $E_{\max}$ and the total number of time slots T is large enough, then the UAV is able to fly to the top of each GU and deliver the video with the minimum link distance (or best channel quality). In this case, it follows from (21) that the maximum transmission rate for GU u_n at time slot t is $R_n^{\max} = B\log_2\left(1+\frac{F^{-1}(\epsilon)P_{\max}\gamma_0}{H^\alpha}\right)$, which is a constant value. Therefore, the total streaming utility of all GUs can be obtained as follows,

$$\begin{aligned} U_{total} &= \delta\theta\sum_{n=1}^{N}\sum_{t=1}^{T} x_n[t]\log\frac{R_n[t](1-\epsilon)}{r_n} \\ &\overset{(g)}{\leq} \delta\theta\sum_{n=1}^{N}\sum_{t=1}^{T} x_n[t]\log\frac{R_n^{\max}(1-\epsilon)}{\min_n r_n} \\ &\overset{(h)}{\leq} \delta\theta T\log\frac{R_n^{\max}(1-\epsilon)}{\min_n r_n} \\ &= \delta\theta T\log\left(\frac{B(1-\epsilon)}{\min_n r_n}\log_2\left(1+\frac{F^{-1}(\epsilon)\gamma_0}{H^\alpha}\right)\right), \end{aligned} \quad (52)$$

where (g) holds due to the definition of $R_n^{\max}$, and (h) holds due to the inequality (11). To achieve performance fairness, the objective in this paper is to maximize the minimum utility among all GUs. Assume that the optimal value of the max-min streaming utility among all GUs is $U^\star$, then $U^\star \leq \delta\theta\sum_{t=1}^{T} x_n[t]\log\frac{R_n[t](1-\epsilon)}{r_n}, \forall n$, and thus $NU^\star \leq U_{total}$. Due to (52), $U^\star \leq \frac{\delta\theta T}{N}\log\left(\frac{B(1-\epsilon)}{\min_n r_n}\log_2\left(1+\frac{F^{-1}(\epsilon)\gamma_0}{H^\alpha}\right)\right)$. Therefore, the upper bound of the max-min streaming utility among all GUs is $\frac{\delta\theta T}{N}\log\left(\frac{B(1-\epsilon)}{\min_n r_n}\log_2\left(1+\frac{F^{-1}(\epsilon)\gamma_0}{H^\alpha}\right)\right)$, which concludes the proof. □

V. Simulation Results

In this section, simulations are conducted and results are provided to evaluate the performance of our proposed solution. For the A2G channel, we adopt the practical Rician fading channel model, which is characterized by the Rician factor K_c. Consider the scenario with $N = 5$ GUs, which are randomly and uniformly located in a geographic area of size 1.6×1.6 km^2. Furthermore, the initial and final locations of the UAV are set as $\mathbf{w}_0 = [-800, 0]^T, \mathbf{w}_F = [800, 0]^T$, respectively. For the parameters in streaming utility function, we set $\theta = 0.8$, and $r_n = 1$ Mbps, $\forall n$, similar to that in [33]. For the proposed Algorithm 1, we set the initial trajectory $\mathbf{Q}^0$ as the direct path from $\mathbf{w}_0$ to $\mathbf{w}_F$ with constant speed $\frac{\|\mathbf{w}_F-\mathbf{w}_0\|}{\delta T}$, and the initial power allocation $\mathbf{P}^0$ is set to be $P[t] = P_{\max}, \forall t$. Furthermore, we assume that $c_1 = 9.26\times10^{-4}$ and $c_2 = 2250$ for the fixed-wing UAV propulsion power consumption parameters [12]. We summarize the parameters used in the simulation in Table II.

TABLE II
Parameters Settings

Parameters	Explanation	Value
H	The altitude of UAV (m)	100 m
$V_{\max}$	The maximum speed of UAV (m/s)	50 m/s
$V_{\min}$	The minimum speed of UAV (m/s)	3 m/s
$a_{\max}$	The maximum acceleration of UAV (m^2/s)	5 m^2/s
$P_{\max}$	Maximum transmit power of UAV (W)	0.1 W
B	Signal bandwidth (Hz)	1 MHz
K_c	Rician fading factor	20
α	Path loss exponent	2
σ^2	Noise power at GUs	-110 dBm
β_0	Channel power gain at the reference distance 1 m	-60 dB
ϵ	Target outage probability threshold	10^{-4}
κ	Algorithm tolerance threshold	10^{-4}
δ	The elemental slot length (s)	1 s

First, we study the convergence of the proposed iterative Algorithm 1. Fig. 2 shows the convergence of Algorithm 1

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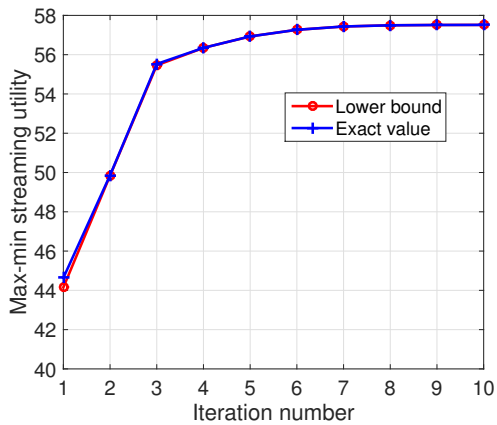
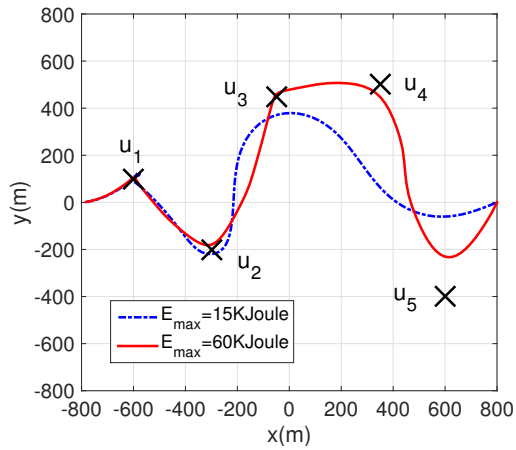


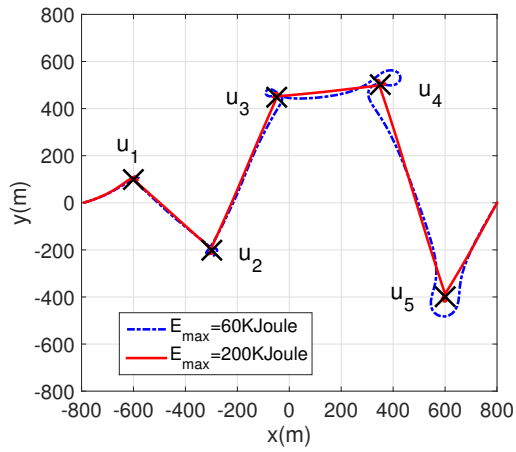
Fig. 2. Convergence of the proposed iterative Algorithm 1.

with $T = 200$ and $E_{\max} = 60$ KJoule. Recall that the optimal solution of (P10) serves as an lower bound to that for (P9) with fixed transmission schedule and transmit power allocation in Algorithm 1. We compare the lower bound, which corresponds to the obtained objective value with Algorithm 1, with the exact value, which refers to the true max-min streaming utility value calculated based on the objective function of (P1). It is observed that the two curves match quite well, which shows that the lower bound for max-min streaming utility with Algorithm 1 is practically tight. Furthermore, it is also observed in Fig. 2 that the proposed algorithm can converge within 8 iterations, which shows the time cost and effectiveness of the proposed solution. It is worthwhile to note that Algorithm 1 is not required to run on the UAV, and can be implemented offline at the workstation on the ground. With given GUs' locations and the intial/final locations of the UAV as well as T, the UAV trajectory $\mathbf{Q}$ (including UAV velocity $\mathbf{V}$ and acceleration $\mathbf{A}$) and transmit power $\mathbf{P}$, as well as transmission schedule $\mathbf{X}$ and transmission rate $\mathbf{R}$ can be obtained by solving problem (P1) with Algorithm 1. After finishing Algorithm 1 and uploading all these results to the UAV, then the UAV can begin its video streaming mission by following the results $\mathbf{Q}, \mathbf{V}, \mathbf{A}, \mathbf{P}, \mathbf{X}, \mathbf{R}$.

The optimized trajectories under different T and energy budget $E_{\max}$ are shown in Fig. 3. It is observed that with the increase of T and energy budget $E_{\max}$, the UAV will move close and hover around each GU to provide better video service. If the energy budget $E_{\max}$ is small, the UAV trajectory mostly consists of paths with relatively large turning radius. However, such a trajectory is expected to incur poor communication channels due to the large distance between UAV and the GU for most of the time. The reason is that the propulsion energy consumption for UAV is the smallest when the UAV flies with speed about 30 m/s due to (1). This is also confirmed by the speed plot in Fig. 4. It is also observed in Fig. 4 that if the energy budget $E_{\max}$ is large enough, the UAV will reduce its flying speed when it is close to each GU, in order to hover around the GU for longer time to obtain better communication channel for video service, as expected.



(a) $T = 100$



(b) $T = 200$

Fig. 3. Optimized trajectories with different T and energy budget $E_{\max}$

Fig. 5 reflects the transmission schedule and transmission rate allocation among all GUs with $T = 200, E_{\max} = 200$ KJoule. It is observed in Fig. 5(a) that the UAV transmits video data to only one GU at each time slot, and the video data is transmitted when the UAV moves closer to the corresponding GU. The transmission rate allocation for all GUs over each time slot is shown in Fig. 5(b). It is observed that the transmission rate varies over time slots and achieves the maximum value for some time slots, since the transmission rate is determined by the time-variant UAV trajectory with a given target outage probability ϵ. Therefore, if the UAV flies closer to the GU, it can increase the transmission rate to achieve the target outage probability ϵ, since the channel quality becomes better. If the UAV hovers around the GU, then the channel quality is the best, and thus the maximum transmission rate for the UAV can be achieved. In Fig. 5(b), it is also observed that the transmission rate varies a little and will not keep identical when UAV hovers around the GU. The reason is that the minimum speed $V_{\min}$ is needed for the fixed-wing UAV to remain aloft and fixed-wing UAV is not able to hover at a fixed location all the time, and thus

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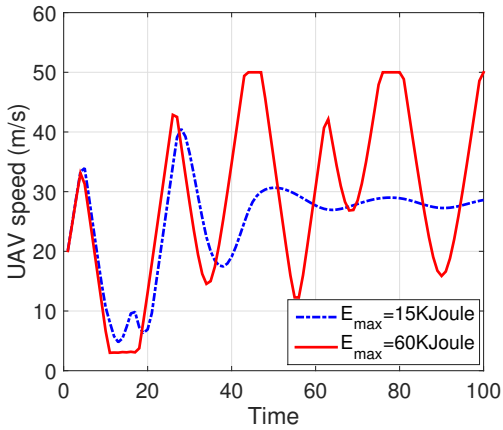
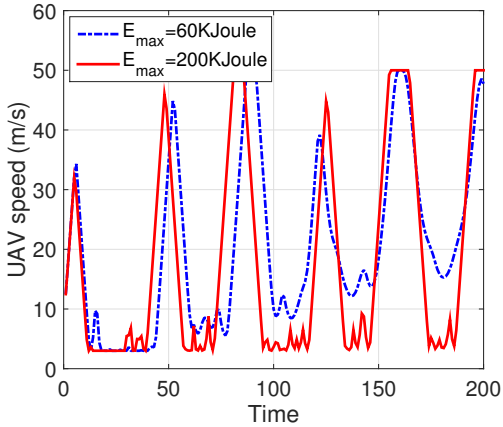
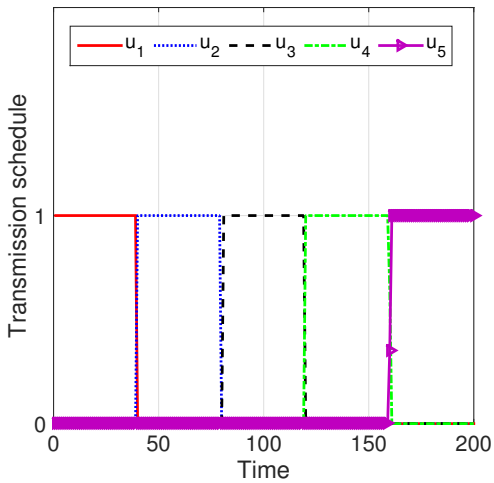


(a) $T = 100$

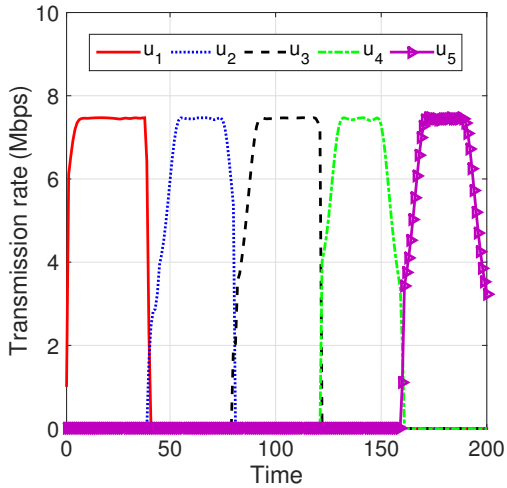


(b) $T = 200$

Fig. 4. UAV speeds with different T and energy budget $E_{\max}$



(a) Transmission schedule among all GUs



(b) Transmission rate allocation among all GUs

Fig. 5. Transmission schedule and transmission rate allocation with $T = 200$, $E_{\max} = 200$ KJoule

the distance between UAV and GU may vary a little when it hovers around the GU.

To show the performance gain of the proposed joint design scheme, we compare the max-min video streaming utility between our proposed solution and three benchmark schemes, namely *Benchmark with resource optimization* and *Benchmark with trajectory optimization*, as well as *Straight flight without optimization*. In the benchmark with resource optimization, the UAV trajectory is fixed as the initial trajectory of Algorithm 1, we only optimize the transmission schedule and transmit power allocation by solving (P6) and (P7) iteratively. In the benchmark with trajectory optimization, the transmission schedule follows a round-robin schedule among all GUs, and the transmit power follows the initial scheme of Algorithm 1, we only optimize the UAV trajectory by solving (P10) iteratively. In the straight flight without optimization, the UAV trajectory and transmit power follow the initial scheme of Algorithm 1, and the transmission schedule follows a round-robin schedule. To show the efficiency of our proposed design, we also compare it with the upper bound obtained in Section IV.

As expected, the proposed solution performs better than the benchmark schemes in terms of the max-min streaming utility, and it is upper bounded by the value which is obtained from theoretical analysis in Section IV. Specifically, it is observed that our proposed solution achieves almost 30% performance gain over benchmark with resource optimization, achieves almost 55% performance gain over benchmark with trajectory optimization, and achieves almost 60% performance gain over straight flight without optimization. The performance gain comes from the joint optimization of UAV trajectory and transmission schedule, as well as the transmit power allocation. In Fig. 6(a), it is observed that the max-min streaming utility increases with the increase of T, and the performance gain is more pronounced when T is large. The reason is that the streaming utility for each GU increases with the increase of T. With our proposed solution, the UAV can hover around each GU to provide a better video streaming service when T is sufficiently large. Fig. 6(b) shows the impact

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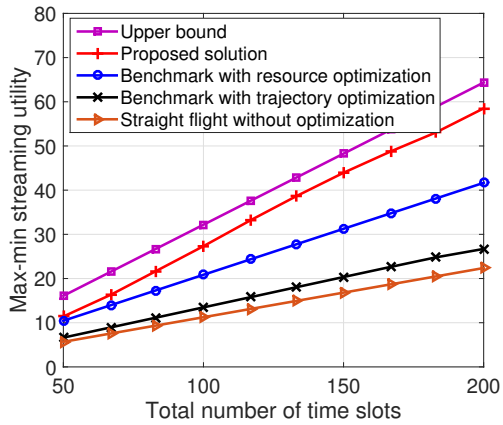
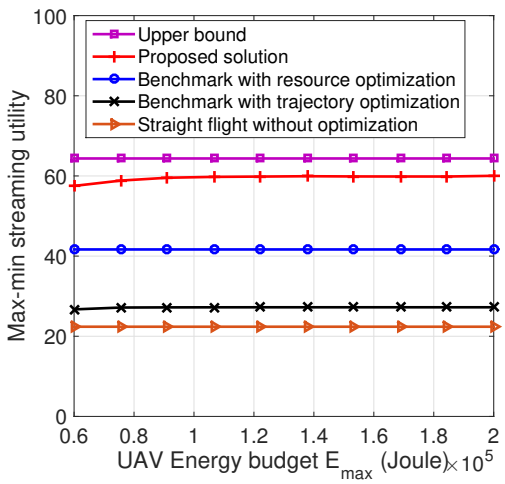
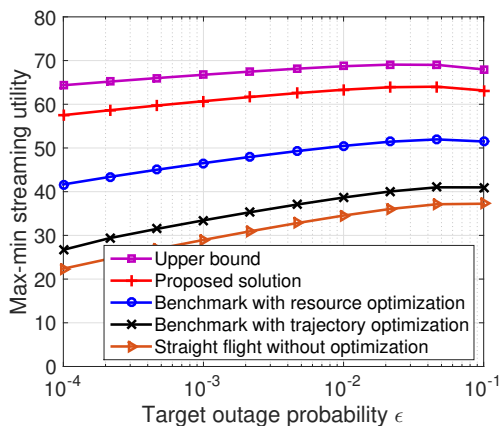


(a) Max-min streaming utility versus T ($\epsilon = 10^{-4}$, $E_{\max} = 60$ KJoule)



(b) Max-min streaming utility versus $E_{\max}$ ($T = 200$, $\epsilon = 10^{-4}$)



(c) Max-min streaming utility versus ϵ ($T = 100$, $E_{\max} = 60$ KJoule)

Fig. 6. Max-min streaming utility versus the time horizon T or UAV energy budget $E_{\max}$ or target outage probability ϵ.

of the UAV energy budget $E_{\max}$ on the max-min streaming utility for different schemes when $T = 200$ and $\epsilon = 10^{-4}$. As expected, it is observed that the streaming utility increases

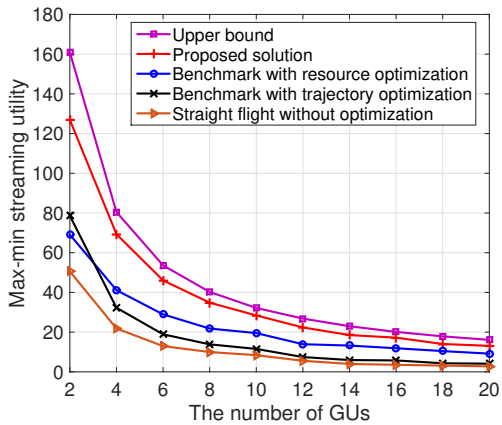


Fig. 7. Average performance with different number of GUs.

with the increase of $E_{\max}$, since the UAV has enough energy to hover over each GU such that it can provide better video service.

In Fig. 6(c), it is observed that the max-min streaming utility increases with the increase of target outage probability ϵ and then decreases when ϵ is large enough, while the performance gain is more pronounced when ϵ is small. The reason is that the transmission rate increases with the increase of ϵ, and thus the streaming utility increases. On the other hand, large outage probability over fading channel will cause video playout to stall and consequently degrade QoE. Such tradeoff is thus shown in Fig. 6(c). Note that only a small performance gap is observed between our proposed design and the upper bound of max-min streaming utility, which reflects that the performance of our proposed solution is quite close to that of the globally optimal solution for the considered setup, which can be obtained by the exhausted search methods for the mixed-integer non-convex programming. To find the global optimal solution efficiently with real situation is non-trivial, and will be left as our future work.

To further show the average performance of the algorithm under different settings, we have conducted simulations with different number of GUs, i.e., N, when $T = 200$, $\epsilon = 10^{-4}$, and $E_{\max} = 60$ KJoule. Given N, the GUs are randomly located in the given area. For each N, the average performance of the algorithm is obtained over 100 independent settings of GUs' locations. It is observed in Fig. 7 that our proposed solution outperforms benchmark schemes and the max-min streaming utility decreases with increase of N. This is expected, since more GUs will compete for the limited communication resources if N is larger.

VI. Conclusion

In this paper, we investigate the adaptive video streaming with multiple users in aerial vehicle networks over fading channel, where the video streaming is adjusted dynamically based on transmission rate. The optimization problem of UAV trajectory design and transmit power allocation, jointly with

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transmission schedule and transmission rate allocation for multiple GUs has been formulated with a non-convex programming. It is revealed that the transmission rate allocation can be transformed to the UAV trajectory design and transmit power allocation with the target outage probability. By applying the SCA and block coordinate descent technique over trajectory design, transmission schedule as well as transmit power allocation, an efficient iteration solution is proposed, which is guaranteed to be locally optimal. Numerical results reveal that the proposed solution can achieve significant gains in terms of the max-min streaming utility. Although the proposed solution is locally optimal and there exist some approximations to model the real situation, our preliminarily work provides a general framework to UAV-enabled video streaming, and can guide the video applications over UAV-enabled wireless systems.

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